

Agent A (Steve)



SENDER-LOCAL FACT
task consequence
carried in REQ

SELF+REQ SCHEMA
SELF: role-local path
REQ: peer request path

*Proposal before resolution;
role-local obligations after resolution*

SELF A: obtain 4 oak_planks → deliver to B
REQ B: craft 1 crafting_table → deposit in order_chest

same initial proposal across paired workcell modes

REQ $r_i \rightarrow_j$

PEER-LOCAL CAPABILITY
B workcell mode

Raw processor
Can transform oak_planks
— vs —
Finished receiver
Can only receive
crafting_table

Agent B (Alex)



peer-local capability

delivered REQ

requested suffix

SHARED EXECUTABLE PRIOR K (shared by both agents)
task structure • skill ontology • typed arguments • grounding constraints

FORWARD RESOLUTION LANE

Receiver learned call
conditioned on
 $x_j \oplus r_i \rightarrow_j$ and K

 $x_j \oplus r_i \rightarrow_j, K \rightarrow u_j$

Canonical resolver
combines proposal q_i^t
with downstream SELF u_j^t

Resolve $\rightarrow (q_i^t, u_j^t)$
 $\rightarrow C_{ij} \rightarrow,^t$

$C_{ij} \rightarrow,^t$
forward-
resolved
commitment

FEEDBACK RESOLUTION LANE

BOUNDED RESPONSE
ACCEPT • REJECT • COUNTER(α)

transparent policy;
no model call

REJECT
 $\rightarrow$ no commitment

ACCEPT /
COUNTER(α)

REQUESTER
RESOLUTION
second learned call
conditioned on
 $q_i \oplus p_j \rightarrow_i$

$C_{ij} \leftarrow,^t$
feedback-
resolved
commitment

RESOLVED SELF+REQ COMMITMENT
 $C \rightarrow$ or $C \leftarrow$
complete role-local obligations

CAPABILITY-CONDITIONED REWRITE same requester view and initial proposal			
Response	Actor	Handoff item	Peer suffix
ACCEPT	B	4 oak_planks	B crafts → deposits
COUNTER(α)	A	1 crafting_table	B deposits only
Destination unchanged: order_chest			

GROUNDED EXECUTION (SHARED CLOSURE)

1 PARSE + CHECK
validated typed
commitment



2 MATERIALIZE + COMPILE
canonical dispatch
→ Mineflayer calls



3 EXECUTE
in-world action
+ handoff



4 VERIFY
inventory +
terminal predicate

